



Bayes Theorem & Bayesian Inference

Lecture 5 - LAS

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Today

- Null Hypothesis Significance Testing
- Bayes Theorem
- Bayesian Inference



Today

[Chapter 7](#) and [Chapter 9](#) from the gitbook.

The information about Bayes theorem, and Bayesian inference is not in the gitbook.



Learning Goals

- Understand and apply Bayes theorem
- Understand the conceptual idea of Bayesian inference
- Can list differences between frequentist and Bayesian inference



Null Hypothesis Significance Testing



Recap NHST (i)

1. Define $\mathcal{H}_0$ and pick α .
2. Assume $\mathcal{H}_0$ is true.
3. Ask: how surprising are our observed data under $\mathcal{H}_0$?
4. $p = P(\text{data at least this extreme} \mid \mathcal{H}_0)$
5. If $p < \alpha$, reject $\mathcal{H}_0$.
6. The sampling distribution tells us how unusual our observed statistic is under $\mathcal{H}_0$
7. The Central Limit Theorem gives a very practical approximation to the sampling distribution



Bechdel-test Data

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	21	957	978
PASS	94	689	783
Total column	115	1646	1761

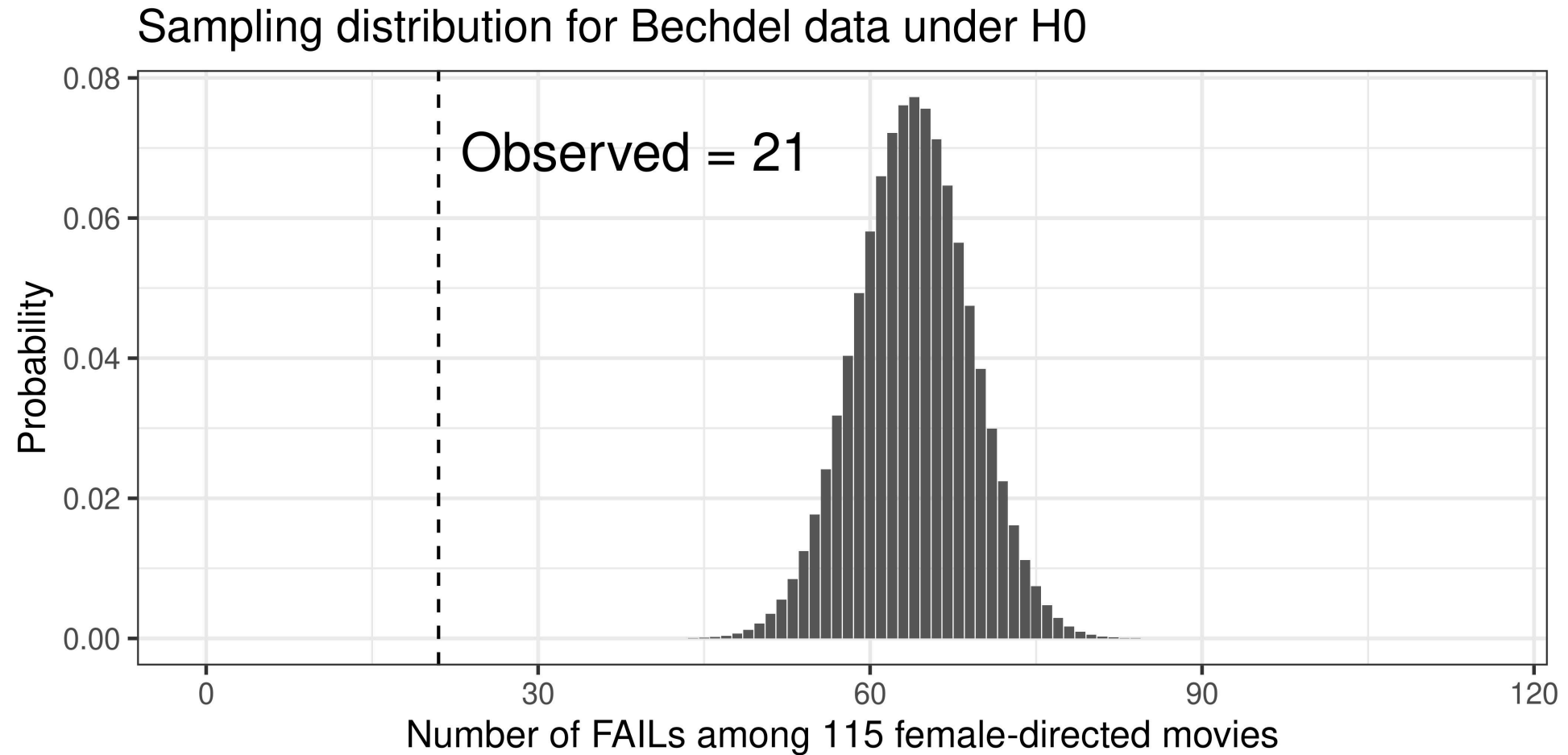


Recap NHST (ii)

1. $\mathcal{H}_0$: Director gender and Bechdel-test outcome are unrelated. $\alpha = 0.05$.
2. Assume $\mathcal{H}_0$ is true.
3. Ask: how surprising are our observed data under $\mathcal{H}_0$?
 - Given, 1761 movies, 115 female-directed and 1646 male-directed, 978 FAIL and 783 PASS,
 - how surprising are 21 FAILs by female directors?
4. $p = P(\text{data at least this extreme} \mid \mathcal{H}_0) = 0.000000000000000000238$
5. If $p < \alpha$, reject $\mathcal{H}_0$.
 - $p < 0.05$, thus reject.



Recap NHST (iii)





Recap NHST (iv)

- We reject or not reject $\mathcal{H}_0$
- Not rejecting $\mathcal{H}_0$ is not evidence for $\mathcal{H}_0$



Recap Central Limit Theorem

Suppose $x_1, x_2, \dots, x_n \stackrel{\text{iid}}{\sim} \mathcal{D}(\mu, \sigma^2, \dots)$.

Then, as n becomes large,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \approx \mathcal{N} \left(\mu, \frac{\sigma^2}{n} \right).$$

Here, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ is known as the *standard error*.



Confidence Intervals

The 95% confidence interval is obtained by:

$$\bar{x} \pm Z_{95\%} \text{SE}_{\bar{x}}$$

where $Z_{95\%}$ is the 97.5th percentile of the standard normal distribution:

$P(Z \leq Z_{95\%}) = 0.975$, so $Z_{95\%} = 1.96$.

Example: $\bar{x} = 1.23$, $\text{SE}_{\bar{x}} = 0.5$, $95\% \text{CI} = [1.23 - 1.96 \times 0.5, 1.23 + 1.96 \times 0.5] = [0.25, 2.21]$.

Definition:

If we repeat *the experiment* infinitely many times and compute a 95% confidence interval each time, then 95% of those intervals will contain the “true” population mean.



Confidence Intervals

The $q\%$ confidence interval is obtained by:

$$\bar{x} \pm Z_{q\%} \text{SE}_{\bar{x}}$$

where $Z_{q\%}$ is

$$P(Z \leq Z_{q\%}) = 1 - \frac{1 - q/100}{2}.$$



Bayes Theorem



Recap Probability Theory

- Joint probability: $P(\text{FAIL} \ \& \ \text{FEMALE})$
- Marginal probability: $P(\text{FAIL})$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE})$



Probability Theory Recap

The Bechdel-test data raw counts:

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	21	957	978
PASS	94	689	783
Total column	115	1646	1761



Probability Theory Recap

The Bechdel-test data as empirical probability distribution:

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	0.012	0.544	0.556
PASS	0.053	0.391	0.444
Total column	0.065	0.935	1.0

- Joint probability: $P(\text{FAIL} \& \text{FEMALE}) = ?$
- Marginal probability: $P(\text{FAIL}) = ?$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE}) = ?$



Probability Theory Recap

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	0.012	0.544	0.556
PASS	0.053	0.391	0.444
Total column	0.065	0.935	1.0

- Joint probability: $P(\text{FAIL} \& \text{FEMALE}) = 0.012$
- Marginal probability: $P(\text{FAIL}) = ?$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE}) = ?$



Probability Theory Recap

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	0.012	0.544	0.556
PASS	0.053	0.391	0.444
Total column	0.065	0.935	1.0

- Joint probability: $P(\text{FAIL} \& \text{FEMALE}) = 0.012$
- Marginal probability: $P(\text{FAIL}) = P(\text{FAIL} \& \text{FEMALE}) + P(\text{FAIL} \& \text{MALE})$
 $= 0.012 + 0.544 = 0.556$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE}) = ?$



Probability Theory Recap

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	0.012	0.544	0.556
PASS	0.053	0.391	0.444
Total column	0.065	0.935	1.0

- Joint probability: $P(\text{FAIL} \ \& \ \text{FEMALE}) = 0.012$
- Marginal probability: $P(\text{FAIL}) = 0.556$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE}) = P(\text{FAIL} \ \& \ \text{FEMALE}) / P(\text{FEMALE}) = 0.012 / 0.065 = 0.185$



Recap Probability Theory

Joint probability: $P(\text{FAIL} \& \text{FEMALE})$

Marginal probability:

$$P(\text{FAIL}) = P(\text{FAIL} \& \text{FEMALE}) + P(\text{FAIL} \& \text{MALE})$$

Conditional probability:

$$\begin{aligned} P(\text{FAIL} \mid \text{FEMALE}) &= \frac{P(\text{FAIL} \& \text{FEMALE})}{P(\text{FEMALE})} \\ &= \frac{P(\text{FAIL} \& \text{FEMALE})}{P(\text{PASS} \& \text{FEMALE}) + P(\text{FAIL} \& \text{FEMALE})} \end{aligned}$$



Probability Theory

Note that:

$$\begin{aligned} P(\text{FAIL} \mid \text{FEMALE}) &= \frac{P(\text{FAIL} \& \text{FEMALE})}{P(\text{FEMALE})} \\ P(\text{FAIL} \mid \text{FEMALE}) \times P(\text{FEMALE}) &= P(\text{FAIL} \& \text{FEMALE}) \\ P(\text{FEMALE} \mid \text{FAIL}) \times P(\text{FAIL}) &= P(\text{FAIL} \& \text{FEMALE}) \end{aligned}$$

Therefore,

$$P(\text{FAIL} \mid \text{FEMALE}) \times P(\text{FEMALE}) = P(\text{FEMALE} \mid \text{FAIL}) \times P(\text{FAIL})$$

and

$$P(\text{FAIL} \mid \text{FEMALE}) = \frac{P(\text{FEMALE} \mid \text{FAIL})P(\text{FAIL})}{P(\text{FEMALE})}$$

Which is Bayes Theorem



Bayes Theorem

DIRECTOR GENDER			
Test	Female	Male	Total row
FAIL	0.012	0.544	0.556
PASS	0.053	0.391	0.444
Total column	0.065	0.935	1.0

- Marginal probability: $P(\text{FEMALE}) = 0.065$
- Conditional probability: $P(\text{FAIL} \mid \text{FEMALE}) = P(\text{FAIL} \& \text{FEMALE}) / P(\text{FEMALE}) = 0.012 / 0.065 = 0.185$
- Marginal probability: $P(\text{FAIL}) = 0.556$
- Conditional probability: $P(\text{FEMALE} \mid \text{FAIL}) = P(\text{FAIL} \& \text{FEMALE}) / P(\text{FAIL}) = 0.012 / 0.556 = 0.023$



Bayes Theorem

$$\begin{aligned}P(\text{FAIL} \mid \text{FEMALE}) \times P(\text{FEMALE}) &= P(\text{FEMALE} \mid \text{FAIL}) \times P(\text{FAIL}) \\(0.012/0.065) \times 0.065 &= (0.012/0.556) \times 0.556 \\0.012 &= 0.012\end{aligned}$$

Or,

$$\begin{aligned}\frac{P(\text{FAIL} \& \text{FEMALE})}{P(\text{FEMALE})} \times P(\text{FEMALE}) &= \frac{P(\text{FAIL} \& \text{FEMALE})}{P(\text{FAIL})} \times P(\text{FAIL}) \\\frac{P(\text{FAIL} \& \text{FEMALE})}{\cancel{P(\text{FEMALE})}} \times P(\cancel{\text{FEMALE}}) &= \frac{P(\text{FAIL} \& \text{FEMALE})}{\cancel{P(\text{FAIL})}} \times P(\cancel{\text{FAIL}}) \\P(\text{FAIL} \& \text{FEMALE}) &= P(\text{FAIL} \& \text{FEMALE})\end{aligned}$$



Bayes Theorem

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$

- In the Bechdel-test data we *knew the entire joint distribution*, hence we didn't really need Bayes theorem.
- In real life we often know some probabilities, but want to infer another.



Bayes Theorem – Why do we need it?

Suppose you take a medical test.

- Outcome: positive (+) or negative (—)

The test comes back *positive*.

What we want to know is:

$$P(D \mid +)$$

Given a positive test, what is the probability that I have the disease?



Bayes Theorem – Medical Tests

- 1% of people have the disease:
 - $P(D) = 0.01$
- If you have the disease, the test is positive 90% of the time:
 - $P(+ \mid D) = 0.90$
- If you do not have the disease, the test is positive 5% of the time:
 - $P(+ \mid \text{not } D) = 0.05$

But we want:

$$P(D \mid +)$$

This is not the same as $P(+ \mid D)$!



Bayes Theorem – Medical Tests

Imagine 10,000 people.

- 1% have the disease $\rightarrow$ 100 people
- 99% do not $\rightarrow$ 9,900 people

Of the 100 people with the disease:

- 90% test positive $\rightarrow$ **90 positive tests**

Of the 9,900 people without the disease:

- 5% test positive $\rightarrow$ **495 positive tests**



Bayes Theorem – Medical Tests

So there are:

- 90 positive tests from people **with** the disease
- 495 positive tests from people **without** the disease

Therefore:

$$P(D \mid +) = \frac{90}{90 + 495} \approx 0.154$$

Even though $P(+ \mid D) = 0.90$. The direction of the conditioning matters!



Bayes Theorem – Medical Tests

We just calculated:

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}$$

where

$$P(+) = P(+ \mid D)P(D) + P(+ \mid \text{not } D)P(\text{not } D).$$

So:

$$P(D \mid +) = \frac{0.90 \times 0.01}{(0.90 \times 0.01) + (0.05 \times 0.99)} \approx 0.154.$$



Summary

- In NHST we can reject $\mathcal{H}_0$, but failing to reject it is not evidence for $\mathcal{H}_0$
- Conditional probabilities are directional: $P(A \mid B) \neq P(B \mid A)$
- Bayes theorem lets us reverse the conditioning.

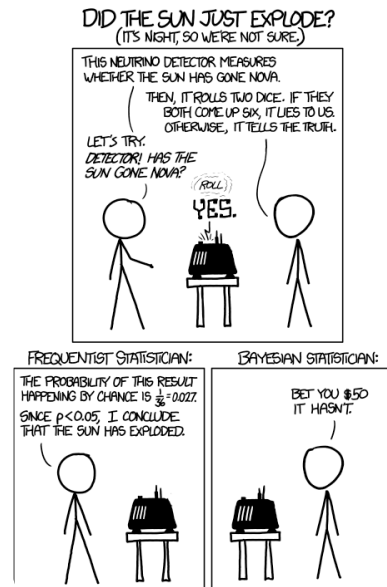


Break

See you in ~10 minutes



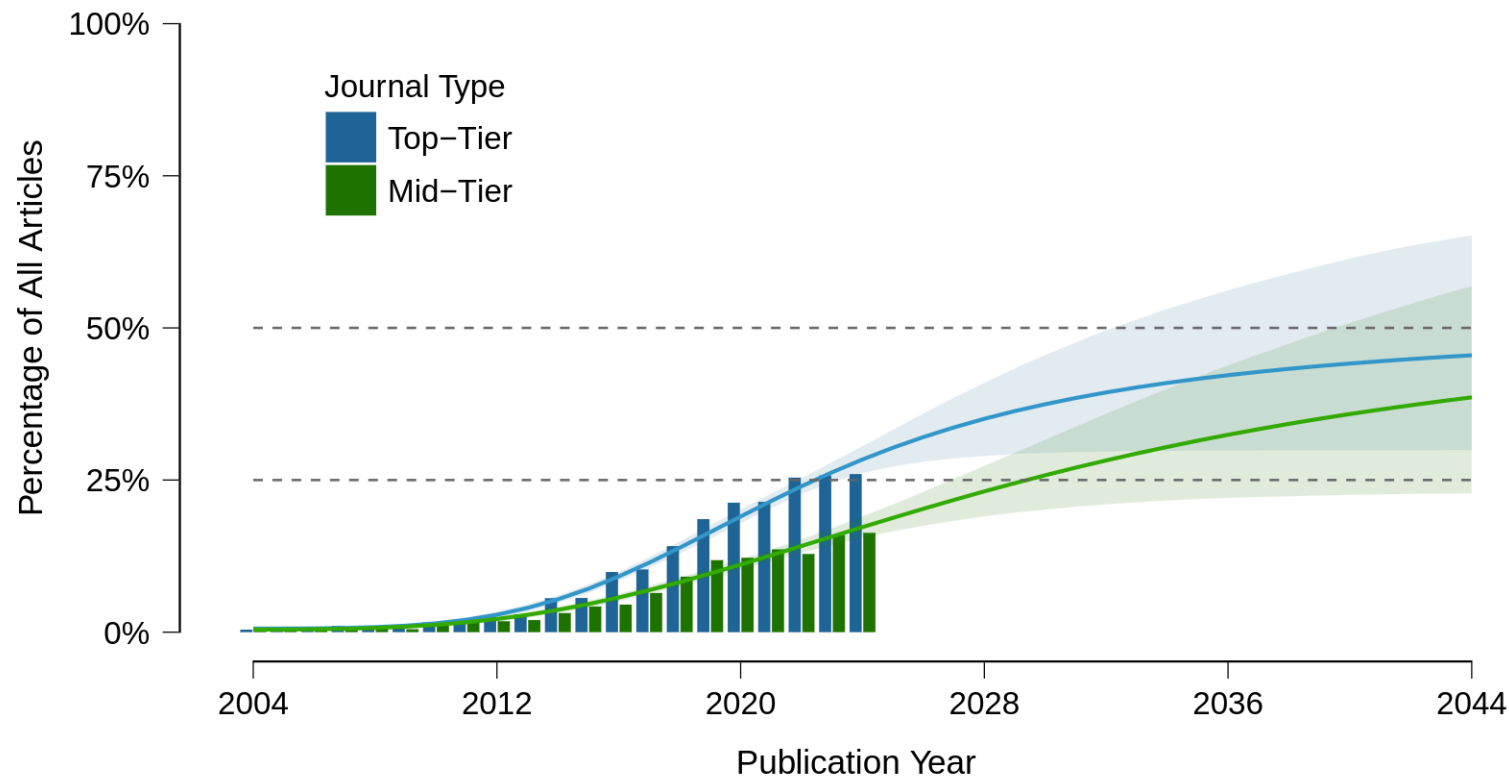
Bayesian Inference





Bayesian Inference is on the Rise

*Observed and Predicted Percentage of **Bayesian** Articles Across Six Top-Tier and Six Mid-Tier Psychology Journals from 2004–2044*





Defining probability (Lecture 2)

Probability: How likely something is to occur.

- Specifically, the proportion of times we expect to observe an outcome in a *random experiment* that could be repeated many times



Bayesian Perspective

For a frequentist, probability describes the long-run frequency of outcomes in repeated random experiments.

For a Bayesian, probability describes *uncertainty about something unknown*.

For example:

- What is the probability that $\mu > 0$?
- What is the probability that $\mathcal{H}_1$ is “true”?



Bayes Theorem Revisited

$$\underbrace{P(\theta \mid \text{data})}_{\text{Posterior Distribution}} = \frac{\overbrace{P(\text{data} \mid \theta)}^{\text{Likelihood}} \times \overbrace{P(\theta)}^{\text{Prior Distribution}}}{\underbrace{P(\text{data})}_{\text{Evidence}}}$$

- The same Bayes theorem, now applied to an unknown parameter θ
- The prior distribution represents what we know about the world *before* seeing the data
 - Is the mean likely near 0?
 - Is a variance between 1 and 2 more likely than 2 and 3?
- Then we observe the data and *update* the prior to a posterior distribution using the likelihood.



Bayesian Updating

In the medical-test example:

$$P(D) = 0.01 \quad \longrightarrow \quad P(D \mid +) = 0.154$$

Before we saw the test the probability that you had the disease was $P(D) = 0.01$.

The positive test *updated* our uncertainty about disease.

Bayesian inference does the same thing for parameters of a population model.



Bayesian Updating

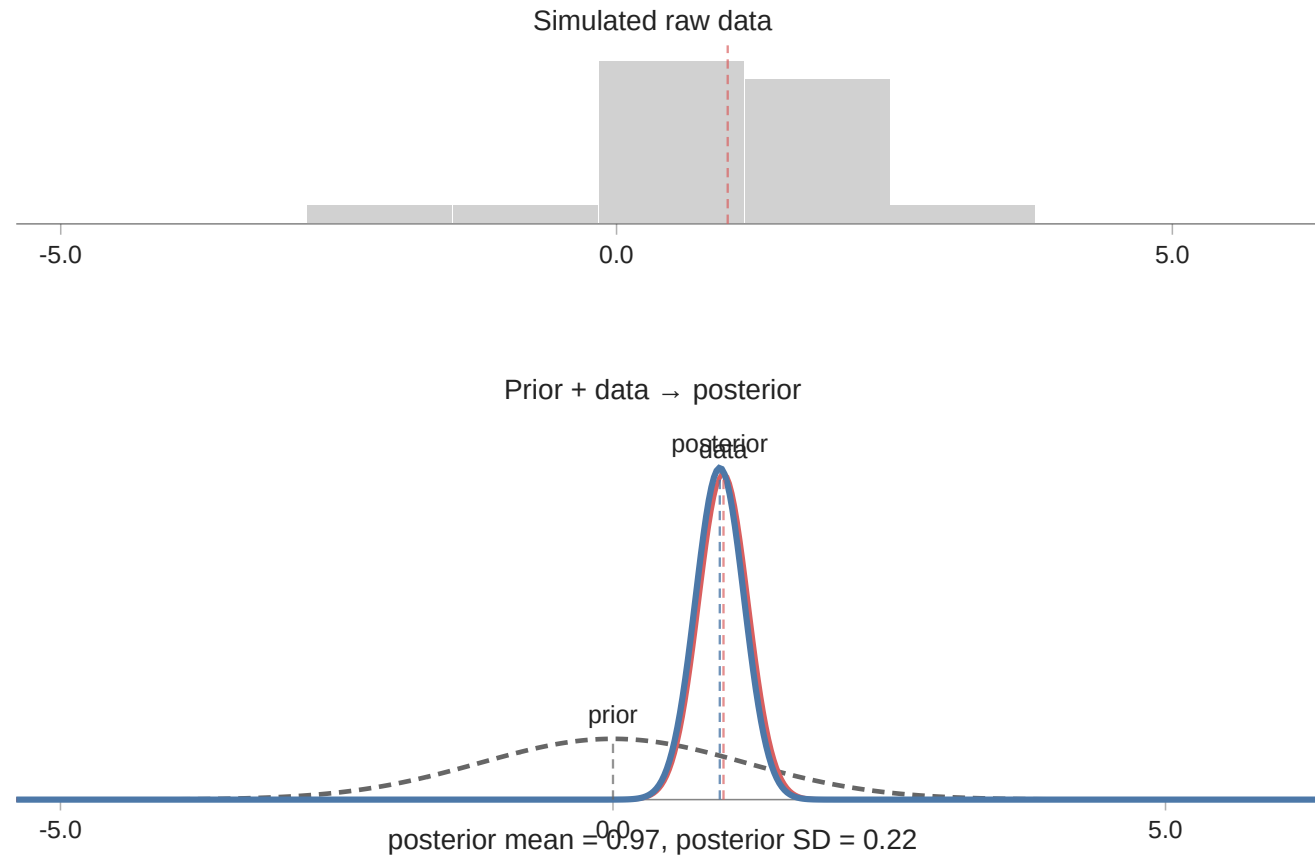
We're learning the population mean and use the prior $\mu \sim \mathcal{N}(m_0, s_0)$.

m_0 0.0

s_0 1.2

$\bar{x}$ 1.0

n 20





Bayes Updating Summary

- The data overwhelm the prior as the sample size grows
- A less dispersed prior requires more data to be overwhelmed
- When the prior and data disagree, the posterior lies between them



Bayesian Credible Intervals

Confidence Interval:

If we repeat *the experiment* infinitely many times and compute a 95% confidence interval each time, then 95% of those intervals will contain the “true” population mean.

Posterior Credible Interval:

After observing the data, there is 95% posterior probability that the population mean lies inside the credible interval.

- Unlike the confidence interval, this statement directly assigns probability to the unknown population mean.
- Unlike the confidence interval, the credible interval says nothing about the “frequency”.



Bayesian Hypothesis Testing

$$P(\mathcal{H} \mid \text{data}) = \frac{P(\text{data} \mid \mathcal{H}) \times P(\mathcal{H})}{P(\text{data})}$$

do this for $\mathcal{H}_0$ and $\mathcal{H}_1$ and obtain $P(\mathcal{H}_0 \mid \text{data})$ and $P(\mathcal{H}_1 \mid \text{data})$

But what is $P(\mathcal{H})$?

A probability distribution over hypotheses.

A frequentist would say, this is nonsense!

$P(\mathcal{H})$ does not exist. A hypothesis is true or false.

A Bayesian argues, $P(\mathcal{H})$ is my subjective belief before I have seen the data about which hypothesis is “true”.



Bayesian Hypothesis Testing

$$P(\mathcal{H} \mid \text{data}) = \frac{P(\text{data} \mid \mathcal{H}) \times P(\mathcal{H})}{P(\text{data})}$$

Then:

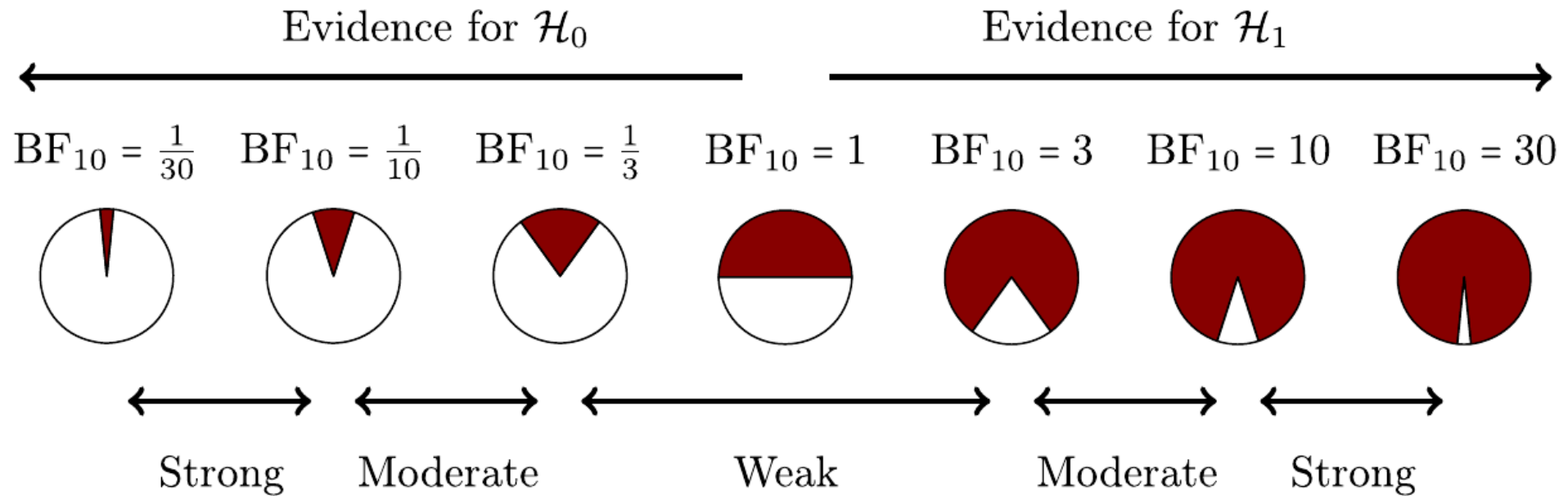
$$\underbrace{\frac{P(\mathcal{H}_1 \mid \text{data})}{P(\mathcal{H}_0 \mid \text{data})}}_{\text{Posterior Odds}} = \underbrace{\frac{P(\text{data} \mid \mathcal{H}_1)}{P(\text{data} \mid \mathcal{H}_0)}}_{\text{Bayes Factor}_{10}} \times \underbrace{\frac{P(\mathcal{H}_1)}{P(\mathcal{H}_0)}}_{\text{Prior Odds}}$$

The Bayes factor (BF) is the “update” from prior to posterior odds.

The $_{10}$ in BF_{10} means $\mathcal{H}_1$ over $\mathcal{H}_0$.



Bayes Factor



We have evidence for $\mathcal{H}_1$, inconclusive evidence, and evidence for $\mathcal{H}_0$!

This is at odds with NHST

Note that $BF_{10} = 1/BF_{01}$, so $BF_{10} = 1/30$ means $BF_{01} = 30$.



Bayes Factor - Summary

NHST	BAYES FACTOR
$P(\text{data} \mid \mathcal{H}_0)$	Compares $P(\text{data} \mid \mathcal{H}_1)$ and $P(\text{data} \mid \mathcal{H}_0)$
Reject or do not reject $\mathcal{H}_0$	Evidence can favor either hypothesis
No evidence <i>for</i> $\mathcal{H}_0$ from a nonsignificant result	Can quantify evidence for $\mathcal{H}_0$
Error rate α	Evidence is expressed as a relative likelihood of the data, no error rate
Binary decision: reject / do not reject	Strength and direction of evidence are continuous